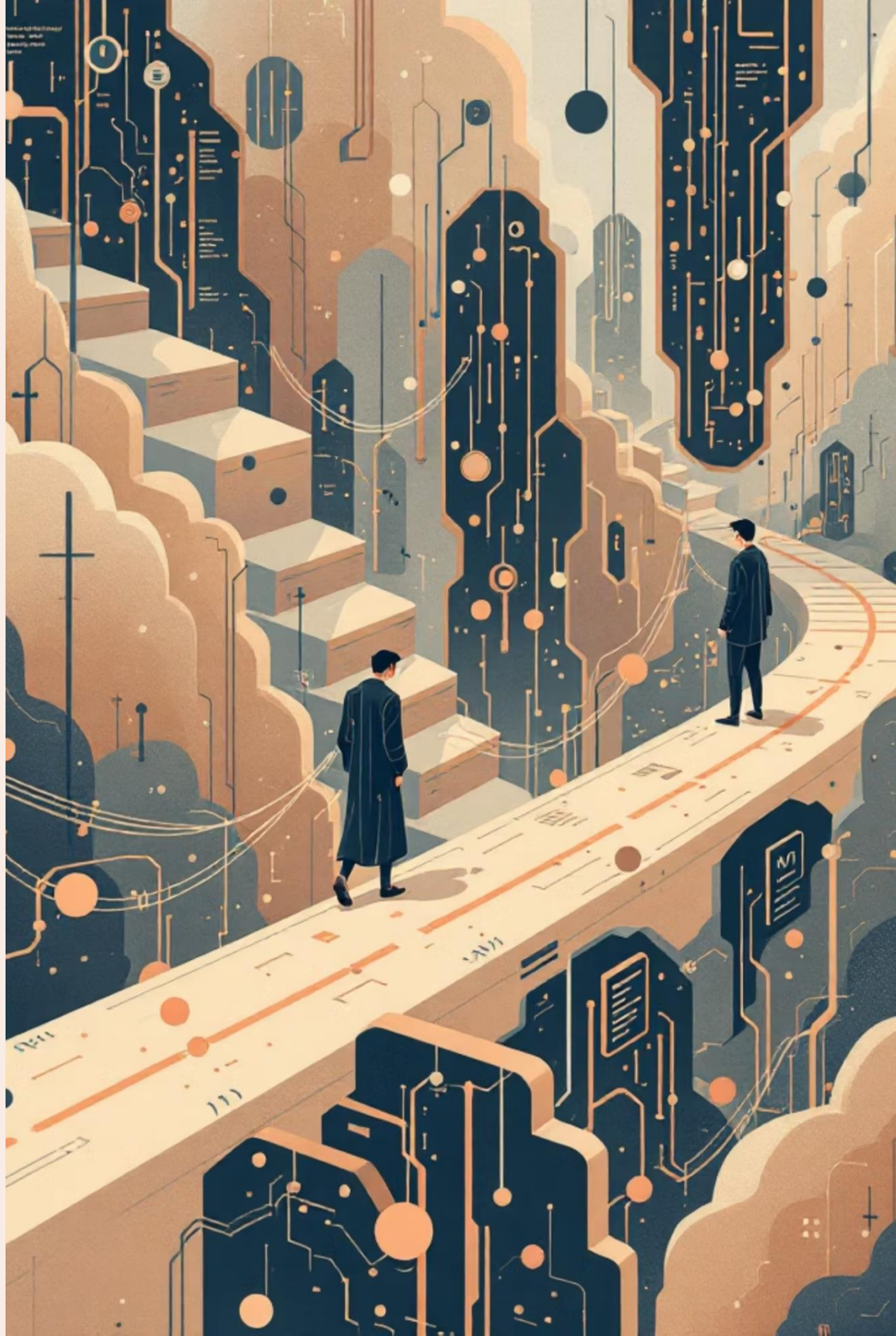
Fine-Tuning Mistral-7B for Text-to-SQL Using QLoRA

CSYE 7380 · Theory and Practical Applications of AI Generative Modeling

Spring 2026 · Northeastern University



Project Overview

This project demonstrates how to fine-tune Mistral-7B-Instruct-v0.2 to convert natural language questions and database schemas into correct SQL queries with plain-English explanations. The approach leverages QLoRA quantization to make training feasible on limited hardware resources.

Base Model: Mistral-7B-Instruct-v0.2

Dataset: gretelai/synthetic_text_to_sql — 100,000 train / 5,851 test samples across 100 domains

Training Subset: 1,000 samples

Hardware: Google Colab Free Tier — Tesla T4 GPU (15.6 GB)

Pipeline: Mistral-7B Base → QLoRA Fine-Tuning → Gradio Deployment

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Technical Configuration — QLoRA

QLoRA combines 4-bit quantization with LoRA adapters, reducing the model from ~14 GB to 4.0 GB and updating only 0.36% of parameters — making a 7B model trainable on free hardware.

4-Bit Quantization

- Quantization type: NormalFloat4 (nf4)
- Compute dtype: bfloat16
- Double quantization: enabled
- Model size after quantization: 4.0 GB

LoRA Configuration

- Rank (r): 16 · Alpha: 32 · Dropout: 0.05
- Target modules: q_proj, k_proj, v_proj, o_proj
- Trainable parameters: 13.6M / 3.77B = 0.36%

Training Arguments

- 1 epoch · 125 steps · Learning rate: 2e-4
- Optimizer: paged_adamw_8bit · Precision: bf16
- Effective batch size: 8 · Training time: ~78 minutes

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Prompt Format & Training Approach

The model is trained on instruction-output pairs using SFTTrainer from the trl library. It learns both SQL correctness and a clean output format simultaneously.

1

Input

Schema + Question

2

Model

Mistral-7B

3

Output

SQL Query + Explanation

Template Used

```
[INST] You are a helpful SQL assistant. Given the database schema
and question, generate the correct SQL query and explain it.

Schema: {sql_context}
Question: {sql_prompt}

[/INST]
SQL Query: {sql}
Explanation: {sql_explanation}
```

Dataset Fields

- sql_context (schema)
- sql_prompt (question)
- sql (ground truth)
- sql_explanation (explanation)

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Training Results — Loss Curve

The model converged steadily across 125 steps in a single epoch.

1.17	0.36	69%
Initial Loss	**Final Loss**	**Reduction**
At step 10	At step 120	Overall improvement

The steepest drop happens in the first 20 steps (1.17 → 0.49), after which the model converges gradually. This pattern is typical for instruction fine-tuning on small, high-quality subsets.

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Before vs. After Fine-Tuning

Example 1 — Avg explainability score in Europe & North America?

Expected: SELECT AVG(explainability_score) FROM creative_ai WHERE region IN ('Europe', 'North America');

Before: ❌ Used GROUP BY region · multiple query attempts · verbose markdown output

After: ✅ Exact match — correct query, clean SQL Query + Explanation format

Example 2 — Decentralized app downloads from Asia-Pacific?

Expected: SELECT SUM(dapp_downloads) FROM dapp_ranking WHERE dapp_region = 'Asia-Pacific';

Before: ❌ Used COUNT() instead of SUM() · wrapped in markdown code blocks

After: ⚠️ Improved — format dramatically cleaner, COUNT(*) vs SUM still off

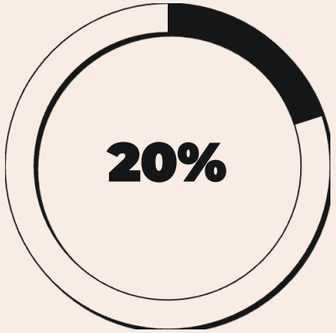
Example 3 — Patients in India who received group therapy?

Expected: SELECT COUNT(*) FROM patients INNER JOIN treatments ON patients.id = treatments.patient_id WHERE treatment = 'Group Therapy' AND country = 'India';

Before: ❌ Rambling explanation · SQL query cut off mid-generation

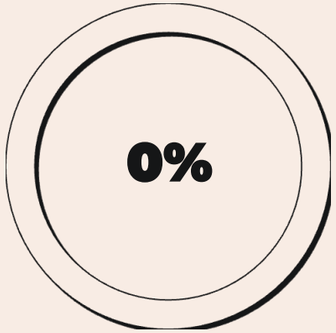
After: ✅ Exact match — correct JOIN and WHERE clause, clean concise output

Evaluation on 20 Test Samples



Exact Match Accuracy

Format Consistency



Cut-off Outputs

Note: Exact match is a strict metric. Qualitative improvement is significantly larger than 20% suggests.

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Gradio Interface & Conclusions

The fine-tuned model is deployed as a Gradio web app — two inputs (schema + question), one output (SQL + explanation), with 3 pre-built demo examples. Launched with share=True for a public 72-hour link.

All 3 Live Demo Examples Produce Correct SQL:

Avg score of students from New York

Correct JOIN + WHERE ✅

Total amount spent per customer

Correct SUM + GROUP BY ✅

Avg salary per department

Correct AVG + GROUP BY ✅

Key Takeaways

- QLoRA makes 7B fine-tuning feasible on free hardware in under 80 minutes
- 1,000 samples (LIMA hypothesis) sufficient for strong format and structural gains
- Exact match understates improvement — format consistency hit 100%, cut-off outputs eliminated
- Full Pre-training → SFT → Deployment pipeline demonstrated end-to-end

Limitations & Future Work

- Only 1 epoch — more training could improve exact match accuracy
- Full 100K dataset would likely yield larger gains
- Execution-based evaluation would better capture real-world correctness

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🔍 Text-to-SQL — Fine-tuned Mistral-7B

Enter a database schema and ask a question in plain English. The fine-tuned model will generate the SQL query and explain it.

Database Schema

```
CREATE TABLE employees (id INT, name TEXT, department TEXT, salary REAL);
```

Your Question

What is the average salary of employees in each department?

Generated SQL + Explanation

SQL Query:
SELECT department, AVG(salary) FROM employees GROUP BY department;

Explanation:
This query calculates the average salary of employees in each department by grouping the data by department and then calculating the average salary for each group.

Clear | Submit | Flag

Examples

Database Schema	Your Question
CREATE TABLE employees (id INT, name TEXT, department TEXT, ...	What is the average salary of employees in each department?
CREATE TABLE orders (order_id INT, customer_id INT, product ...	Find the total amount spent by each customer in 2024
CREATE TABLE students (id INT, name TEXT, grade INT, city TE...	What is the average score of students from New York?

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Enter a database schema and ask a question in plain English. The fine-tuned model will generate the SQL query and explain it.

Database Schema

```
CREATE TABLE orders (order_id INT, customer_id INT, product TEXT, amount REAL, order_date DATE); CREATE TABLE customers (customer_id INT, name TEXT, city TEXT);
```

Your Question

Find the total amount spent by each customer along with their name

Generated SQL + Explanation

SQL Query:
SELECT c.name, SUM(o.amount) as total_spent FROM orders o JOIN customers c ON o.customer_id = c.customer_id GROUP BY c.name;

Explanation:
This query joins the orders and customers tables on the customer_id column, groups the results by the name column, and calculates the total amount spent by each customer.

Clear | Submit | Flag

Examples

Database Schema	Your Question
CREATE TABLE employees (id INT, name TEXT, department TEXT, ...	What is the average salary of employees in each department?
CREATE TABLE orders (order_id INT, customer_id INT, product ...	Find the total amount spent by each customer in 2024
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Enter a database schema and ask a question in plain English. The fine-tuned model will generate the SQL query and explain it.

Database Schema

```
CREATE TABLE students (id INT, name TEXT, grade INT, city TEXT); CREATE TABLE scores (student_id INT, subject TEXT, score INT);
```

Your Question

What is the average score of students from New York?

Generated SQL + Explanation

SQL Query:
SELECT AVG(s.score) FROM scores s JOIN students st ON s.student_id = st.id WHERE st.city = 'New York';

Explanation:
This query calculates the average score of students from New York by joining the scores and students tables on the student_id column and filtering for students from New York.

Clear | Submit | Flag

Examples

Database Schema	Your Question
CREATE TABLE employees (id INT, name TEXT, department TEXT, ...	What is the average salary of employees in each department?
CREATE TABLE orders (order_id INT, customer_id INT, product ...	Find the total amount spent by each customer in 2024
CREATE TABLE students (id INT, name TEXT, grade INT, city TE...	What is the average score of students from New York?

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Enter a database schema and ask a question in plain English. The fine-tuned model will generate the SQL query and explain it.

Database Schema

```
CREATE TABLE employees (id INT, name TEXT, department TEXT, salary INT);
```

Your Question

Show me all employees in the sales department

Generated SQL + Explanation

Clear | Submit | Flag

Examples

Database Schema	Your Question
CREATE TABLE employees (id INT, name TEXT, department TEXT, ...	What is the average salary of employees in each department?
CREATE TABLE orders (order_id INT, customer_id INT, product ...	Find the total amount spent by each customer in 2024
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